Simplify fractions

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Simplify 42/56. Copy or complete the first step.

2. Which is greater, 7/12 or 5/8?

3. Order 2/3, 3/4 and 5/6 from least to greatest.

4. Miro says: Miro compares fractions by looking only at the denominators. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Simplify 42/56. Copy or complete the first step.
   Answer 3/4.

2. Which is greater, 7/12 or 5/8?
   Answer 5/8 because 7/12 = 14/24 and 5/8 = 15/24.

3. Order 2/3, 3/4 and 5/6 from least to greatest.
   Answer 2/3, 3/4, 5/6.

4. Miro says: Miro compares fractions by looking only at the denominators. Is this correct? Explain.
   Answer Convert the fractions to equivalent forms or use a shared benchmark before comparing.

Simplify fractions

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ______________

1. Simplify 42/56.

2. Which is greater, 7/12 or 5/8?

3. Order 2/3, 3/4 and 5/6 from least to greatest.

4. Solve this independently and show every stage: Simplify 42/56.

5. Use a second method or representation for: Which is greater, 7/12 or 5/8?

6. Check this result using an inverse, estimate or known fact: Order 2/3, 3/4 and 5/6 from least to greatest.

7. Prove or explain your reasoning: Give a fraction between 3/5 and 2/3.

8. Identify and correct this misconception: Miro compares fractions by looking only at the denominators.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Order 2/3, 3/4 and 5/6 from least to greatest.

Teacher answers and checking prompts

1. Simplify 42/56.
   Answer 3/4.

2. Which is greater, 7/12 or 5/8?
   Answer 5/8 because 7/12 = 14/24 and 5/8 = 15/24.

3. Order 2/3, 3/4 and 5/6 from least to greatest.
   Answer 2/3, 3/4, 5/6.

4. Solve this independently and show every stage: Simplify 42/56.
   Answer 3/4.

5. Use a second method or representation for: Which is greater, 7/12 or 5/8?
   Answer Expected result: 5/8 because 7/12 = 14/24 and 5/8 = 15/24. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Order 2/3, 3/4 and 5/6 from least to greatest.
   Answer Expected result: 2/3, 3/4, 5/6. A valid check must be shown.

7. Prove or explain your reasoning: Give a fraction between 3/5 and 2/3.
   Answer 5/8 is one possible answer. The explanation must show why the method works.

8. Identify and correct this misconception: Miro compares fractions by looking only at the denominators.
   Answer Convert the fractions to equivalent forms or use a shared benchmark before comparing.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Order 2/3, 3/4 and 5/6 from least to greatest.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Simplify fractions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: Give a fraction between 3/5 and 2/3.

2. Prove the answer to this question, rather than only calculating it: Simplify 42/56.

3. Solve this in two different ways and compare them: Which is greater, 7/12 or 5/8?

4. Work backwards from the answer to reconstruct the question: 2/3, 3/4, 5/6.

5. Explain why this reasoning fails: Miro compares fractions by looking only at the denominators.

6. Create a new example that tests the same idea as: Order 2/3, 3/4 and 5/6 from least to greatest.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Give a fraction between 3/5 and 2/3.
   Answer 5/8 is one possible answer. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Simplify 42/56.
   Answer Expected result: 3/4. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Which is greater, 7/12 or 5/8?
   Answer Expected result: 5/8 because 7/12 = 14/24 and 5/8 = 15/24. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 2/3, 3/4, 5/6.
   Answer One valid reconstruction is: Order 2/3, 3/4 and 5/6 from least to greatest. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro compares fractions by looking only at the denominators.
   Answer Convert the fractions to equivalent forms or use a shared benchmark before comparing.

6. Create a new example that tests the same idea as: Order 2/3, 3/4 and 5/6 from least to greatest.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.